

UAV Engine Digital Twin Phase 1A Progress Report

Date: 3 Sep 2026 | Project: SIH – UAV Engine PHM

WHAT WE BUILT TODAY

Phase 1A is COMPLETE – the foundational pipeline that detects engine degradation by comparing real sensor readings against healthy predictions.

THE PIPELINE (3 modules):

```
Sensor Data → Digital Twin → Residual Engine → Health Signal
(12 sensors)  (ML model that predicts healthy values)  (compares actual vs expected, outputs z-scores)
```

MODULES BUILT:

1. simulation/telemetry_generator.py
Simulates realistic UAV engine flights with 12 sensors.
Generates both healthy runs and degrading engine runs.
2. ml/digital_twin.py
XGBoost model trained ONLY on healthy engine data.
Given operating conditions (RPM, throttle, altitude...),
predicts what a healthy engine's sensors should read.
3. backend/residual.py
Calculates difference between actual and healthy predictions.
Normalizes as z-score so all sensors are comparable.

KEY RESULT:

```
z-score near 0 → Engine healthy (actual matches expected)
z-score > 3    → Something is wrong
z-score > 10   → CRITICAL degradation
```

System shows 15-16x difference between healthy and critical.

Generated Data & Model Training

GENERATED DATA (8 files in data/ folder)

Training data (healthy engines):

healthy_run_01.jsonl ... healthy_run_05.jsonl
→ 5,500 samples total (5 flights × 1,100 seconds each)

Test data (degrading engines):

trajectory_run_01.jsonl ... trajectory_run_03.jsonl
→ 3,300 samples (3 flights, healthy → critical)

Each sample has 12 sensors:

INPUTS: rpm, throttle, map, altitude, ambient_temp

OUTPUTS: egt, cht, oil_pressure, oil_temp, vibration, fuel_flow

MODEL TRAINING RESULTS

6 XGBoost regressors trained (one per output sensor):

Sensor	Mean Absolute Error	Quality
EGT	1.77 °C	Excellent
CHT	1.20 °C	Excellent
Oil Pressure	0.60 psi	Excellent
Oil Temp	0.87 °C	Excellent
Vibration	0.017 g	Excellent
Fuel Flow	0.18 L/h	Excellent

ACCEPTANCE TEST (healthy vs critical z-scores)

Sensor	z Healthy	z Critical	Ratio
EGT	1.30	20.13	15.5x
Oil Pressure	1.17	19.02	16.2x
Vibration	1.25	18.12	14.5x

21/21 unit tests PASSING.

How to Run the Demo

Share these commands with the team

PREREQUISITES

- Python 3.10+ installed
- Open terminal in: e:\Project\SIH\UAV

STEP 1: Interactive Demo (shows everything step by step)

```
.venv\Scripts\python.exe demo.py
```

Press Enter to advance through 5 steps.
Shows real data, predictions, residuals, and the plot.

STEP 2: Run All Tests (21/21 should pass)

```
.venv\Scripts\python.exe -m pytest tests/ -v
```

STEP 3: View the Acceptance Plot

Open: tests\acceptance_plot.png

OTHER COMMANDS:

```
# Regenerate synthetic data:  
.venv\Scripts\python.exe -m simulation.telemetry_generator
```

```
# Retrain the model:  
.venv\Scripts\python.exe -m ml.digital_twin
```

```
# Regenerate the acceptance plot:  
.venv\Scripts\python.exe tests/test_acceptance.py
```

WHAT'S NEXT – PHASE 1B

Phase 1A gives us residuals (the signal).
Phase 1B uses residuals to build:

1. Anomaly Detector → flags when something is wrong
2. Fault Classifier → identifies WHAT is wrong
3. Health Index → overall engine health (0-100%)
4. RUL Estimator → remaining hours until failure

Then: Dashboard (Phase 2) and Mission Risk (Phase 3).

Acceptance Test – Visual Proof

Left: Actual (red) vs Expected (blue) | Right: Z-score residuals
Background: green=healthy, yellow=early, orange=incipient, red=progressive, purple=critical

Phase 1A Acceptance Test -- Digital Twin Residuals
(Background color = degradation stage)

